Lab 7: Fuzz Testing

ATM Withdrawal (L6_L7_18_atm_withdrawal)

Name: Shreesh Athreya

SRN: PES1UG24CS441

Class: 5H

Repository: https://github.com/cswizard7/L6_L7_18_atm_withdrawal

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Step 1: Installing dependencies

Commands: `python -m pip install hypothesis` and `python -m pip install pytest`

Installed Hypothesis 6.170.15 and pytest 9.2.0 on Python 3.13.1 (Windows 11).

Step 2: Running the buggy application

Command: `python atm_withdrawal.py`

Entering the non-numeric amount "abc" crashes notes_needed with an unhandled ValueError, because the input is passed straight to int() without validation.

```
C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): abc
Traceback (most recent call last):
  File "C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7\atm_withdrawal.py", line 29, in <module>
    main()
    ~~~~^^
  File "C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7\atm_withdrawal.py", line 15, in main
    result = notes_needed(user_input)
  File "C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7\atm_withdrawal.py", line 2, in notes_needed
    return int(amount.replace(',', '')) // 100
           ~~~^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: 'abc'

C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>
```

Figure 1: Buggy application crashing on invalid input

Step 3: Hypothesis fuzz tests

test_atm_withdrawal.py contains six Hypothesis tests covering all three functions:

- **No-crash tests:** each function is fuzzed with random Unicode text and None (st.text() | st.none()); any exception counts as a failure.
- **Input-validation tests:** generated integers, digit groups and CSV lists check that invalid amounts and card numbers are rejected and valid ones give the correct result.

Step 4: Fuzz tests on the buggy application

Command: `python -m pytest test_atm_withdrawal.py`

All 6 tests fail. Hypothesis found 2 distinct failures in several of the tests and shrank each one to a minimal failing input, such as amounts=[0].

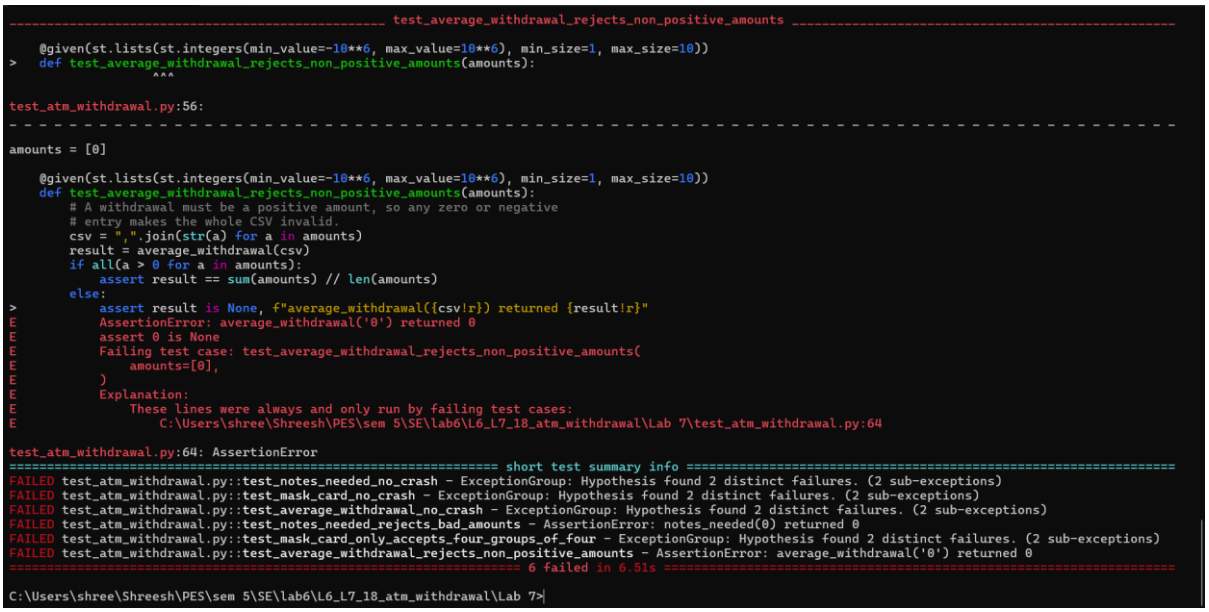


Figure 2: All six fuzz tests failing on the buggy code

Bugs found and fixes applied

Function	Bug found by fuzzing	Fix
`notes_needed`	Crashed on None and non-numeric text (AttributeError / ValueError); returned a note count for 0, negative amounts and amounts not a multiple of 100.	Checks the input is a string of ASCII digits, rejects zero and non-multiples of 100, returns None for invalid input.
`mask_card`	Crashed on None and on fewer than 4 groups (IndexError); accepted any text with 4+ groups, e.g. '1 2 3 4'.	Accepts only exactly 4 groups of 4 ASCII digits; returns None otherwise.
`average_withdrawal`	Crashed on None, empty string, empty entries like '1,,3' and non-numeric text; averaged zero and negative withdrawals.	Validates every entry as a positive whole number; returns None if any entry is invalid.

Step 5: Running the fixed application

Command: `python atm_withdrawal.py` (run twice)

First run with invalid input: each function now prints a clear "Invalid …" message instead of crashing. Second run with valid input: the results are unchanged.

```
C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): abc
Invalid amount: enter a positive multiple of 100

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 1234
Invalid card number: enter 4 groups of 4 digits

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 500,,c
Invalid input: enter positive whole numbers separated by commas

C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): 2,500
Number of 100 Notes: 25

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 1234 5678 9012 3456
Masked Card: XXXX XXXX XXXX 3456

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 500,1000,2000
Average Withdrawal: 1166

C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>
```

Figure 3: Fixed application handling invalid and valid input

Step 6: Fuzz tests on the fixed application

Command: `python -m pytest test_atm_withdrawal.py`

All 6 fuzz tests pass against the fixed code.

```
C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7>python -m pytest test_atm_withdrawal.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.1, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\shree\Shreesh\PES\sem 5\SE\lab6\L6_L7_18_atm_withdrawal\Lab 7
plugins: anyio-4.12.0, hypothesis-6.170.15
collected 6 items

test_atm_withdrawal.py ......                                                                                                      [100%]

============================================================ 6 passed in 0.85s =============================================================
```

Figure 4: All six fuzz tests passing on the fixed code